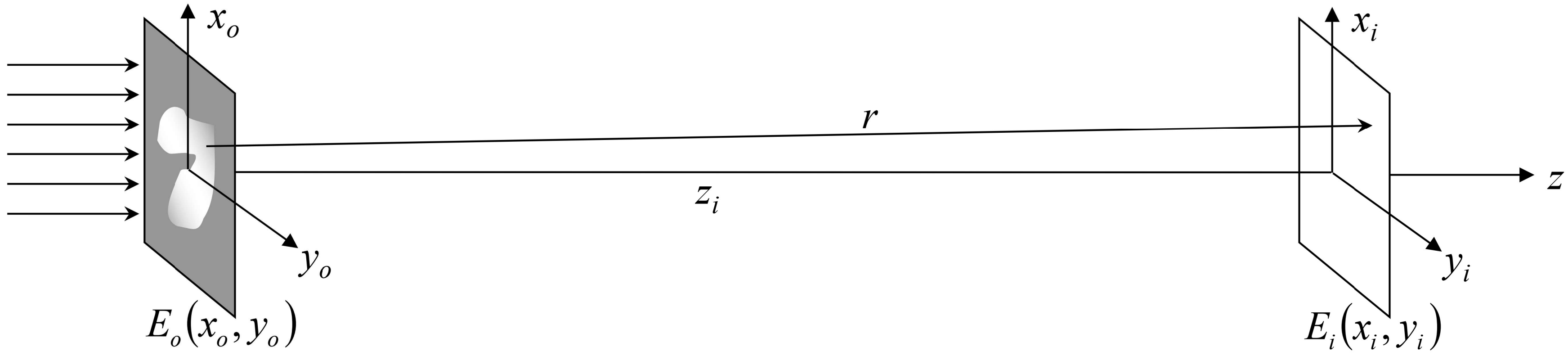
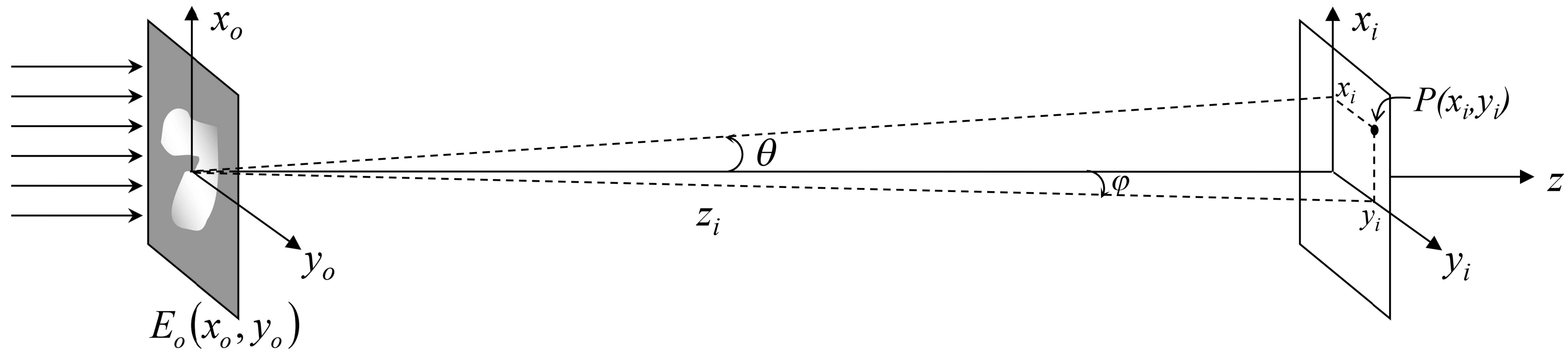
Fresnel-Kirchhoff Diffraction Formula



Far-field diffraction and Spatial Fourier Transform

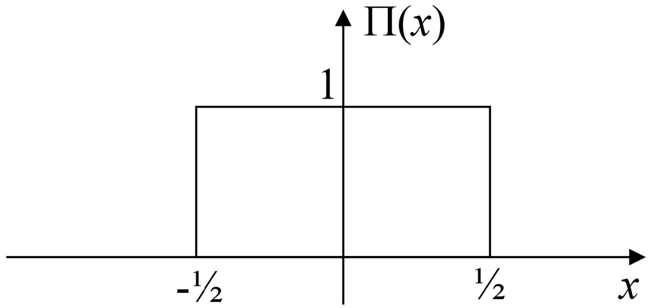
θ and φ are defined as the angles subtended by the point $P(x_i, y_i)$ in the x-z plane and the y-z plane, respectively (see figure below).



Rectangle function and its F.T.

Before proceeding, let’s review a few commonly used functions and their Fourier Transforms.

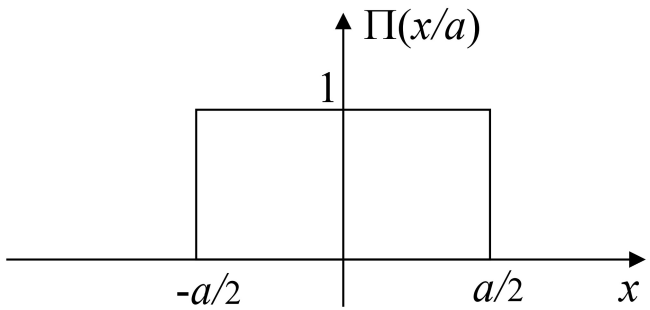
(1) Rectangle function



$$\Pi(x) \equiv \begin{cases} 1 & |x| \le 1/2 \\ 0 & |x| > 1/2 \end{cases} \quad (13\text{-}10)$$

$$\mathcal{F}\{\Pi(x)\} = \frac{\sin(\pi f)}{\pi f} \equiv \mathrm{sinc}\, f \quad (13\text{-}11)$$

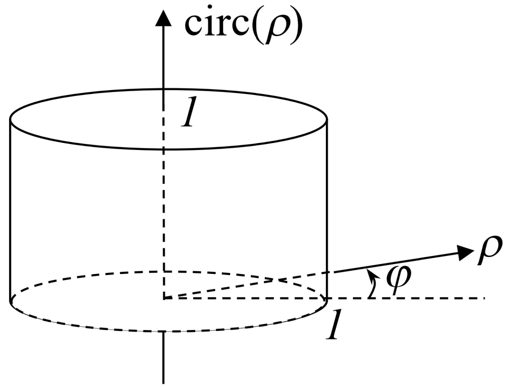
Rectangle function with scaling:



$$\mathcal{F}\left\{\Pi\left(\frac{x}{a}\right)\right\} = a\,\mathrm{sinc}(af) \quad (13\text{-}12)$$

Circle Function and its F.T.

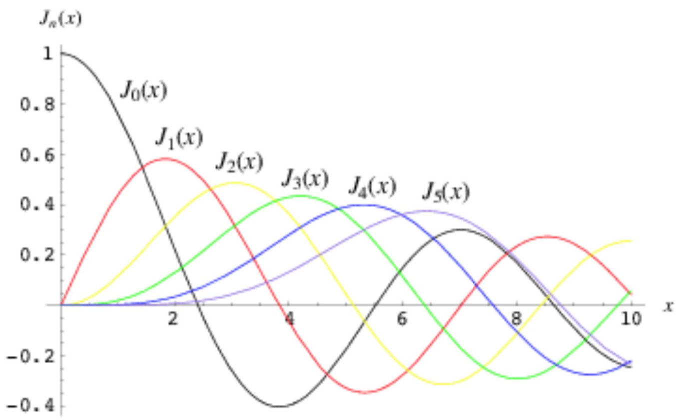
(2) Circle function



$$\mathrm{circ}(\rho) \equiv \begin{cases} 1 & \rho \le 1 \\ 0 & \rho > 1 \end{cases} \qquad (13\text{-}13)$$

$$\mathcal{F}\{\mathrm{circ}(\rho)\} = \frac{\mathrm{J}_1(2\pi f)}{f} \qquad (13\text{-}14)$$

J_1 is the Bessel function of the first kind[1] of order 1.

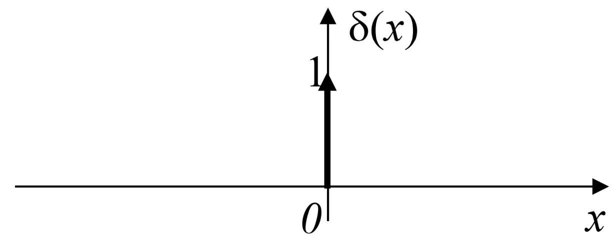


Footnote:

1. The left plot shows the various orders of the Bessel functions of the first kind. They are also called the cylindrical harmonics. Any arbitrary function in the cylindrical coordinates can be expressed as the linear superposition of these harmonics. To find the derivation of Eq (13-14), you can go to: http://mathworld.wolfram.com/CylinderFunction.html

Delta function and its F.T.

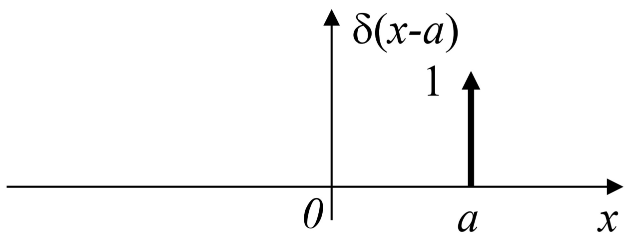
(3) Delta function



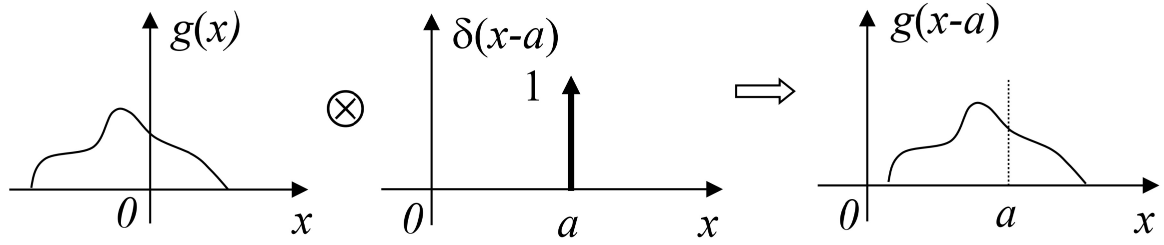
$$\lim_{\varepsilon \to 0} \int_{-\varepsilon}^{+\varepsilon} \delta(x)\, dx = 1 \qquad (13\text{-}15)$$

$$\mathcal{F}\{\delta(x)\} = 1 \qquad (13\text{-}16)$$

Delta function with a shift:



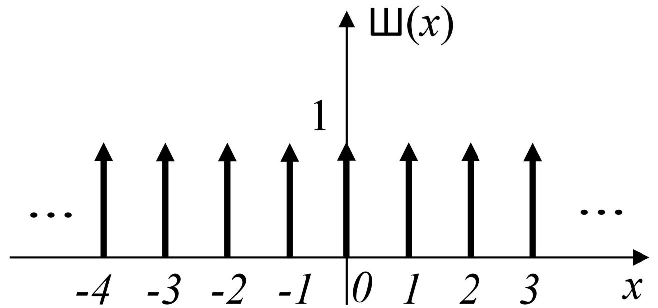
$$\mathcal{F}\{\delta(x-a)\} = e^{-i2\pi fa} \qquad (13\text{-}17)$$



$$\mathcal{F}\{g(x) \otimes \delta(x-a)\} = e^{-i2\pi fa}\, \mathcal{F}\{g(x)\} \qquad (13\text{-}18)$$

Impulse train function and its F.T.

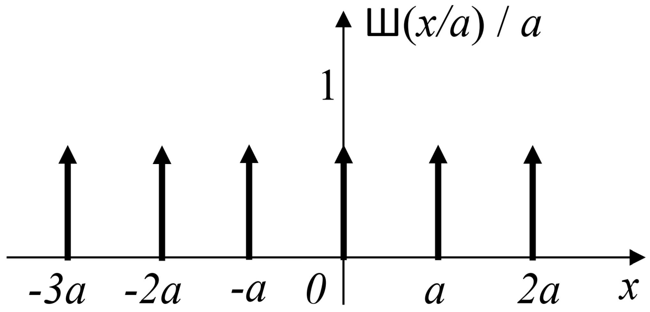
(4) Impulse Train Function (Shah function)



$$\text{Ш}\left(x\right) \equiv \sum_{n=-\infty}^{+\infty} \delta\left(x-n\right) \qquad (13\text{-}19)$$

$$\mathcal{F}\left\{\text{Ш}\left(x\right)\right\} = \text{Ш}\left(f\right) \qquad (13\text{-}20)$$

Shah function with scaling:

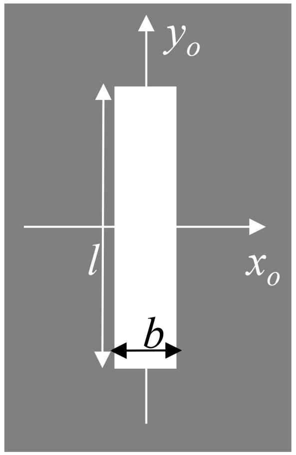


$$\frac{\text{Ш}\left(x/a\right)}{a} \equiv \sum_{n=-\infty}^{+\infty} \delta\left(x-na\right) \qquad (13\text{-}21)$$

$$\mathcal{F}\left\{\frac{\text{Ш}\left(x/a\right)}{a}\right\} = \text{Ш}\left(af\right) \qquad (13\text{-}22)$$

Single-Slit Far-Field Diffraction

Example 13-1: Find and plot the far-field diffraction pattern of a single rectangular slit of dimension $b \times l$.



Aperture function: $$E_o(x_o, y_o) = \Pi\left(\frac{x_o}{b}\right)\Pi\left(\frac{y_o}{l}\right) \quad (13\text{-}23)$$

Far-field E field distribution:

$$E_i(x_i, y_i) \propto \mathcal{F}\{E_o(x_o, y_o)\}\Big|_{f_x=\frac{x_i}{\lambda z_i}, f_y=\frac{y_i}{\lambda z_i}} \quad (13\text{-}24)$$

$$\mathcal{F}\{E_o(x_o, y_o)\} = \mathcal{F}\left\{\Pi\left(\frac{x_o}{b}\right)\Pi\left(\frac{y_o}{l}\right)\right\} = bl\,\mathrm{sinc}(bf_x)\,\mathrm{sinc}(lf_y) \quad (13\text{-}25)$$

Far-field Intensity distribution:

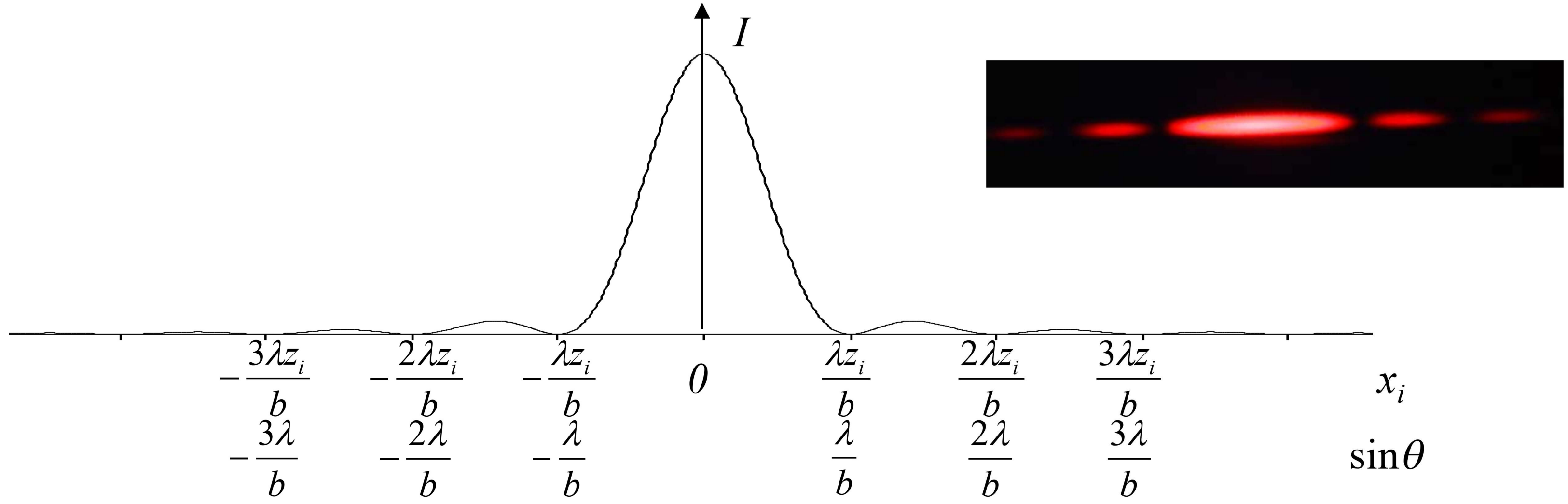
$$I_i(x_i, y_i) \propto E_i^2(x_i, y_i) \propto \mathrm{sinc}^2(bf_x)\,\mathrm{sinc}^2(lf_y)\Big|_{f_x=\frac{x_i}{\lambda z_i}, f_y=\frac{y_i}{\lambda z_i}} = \mathrm{sinc}^2\left(\frac{bx_i}{\lambda z_i}\right)\mathrm{sinc}^2\left(\frac{ly_i}{\lambda z_i}\right) \quad (13\text{-}26)$$

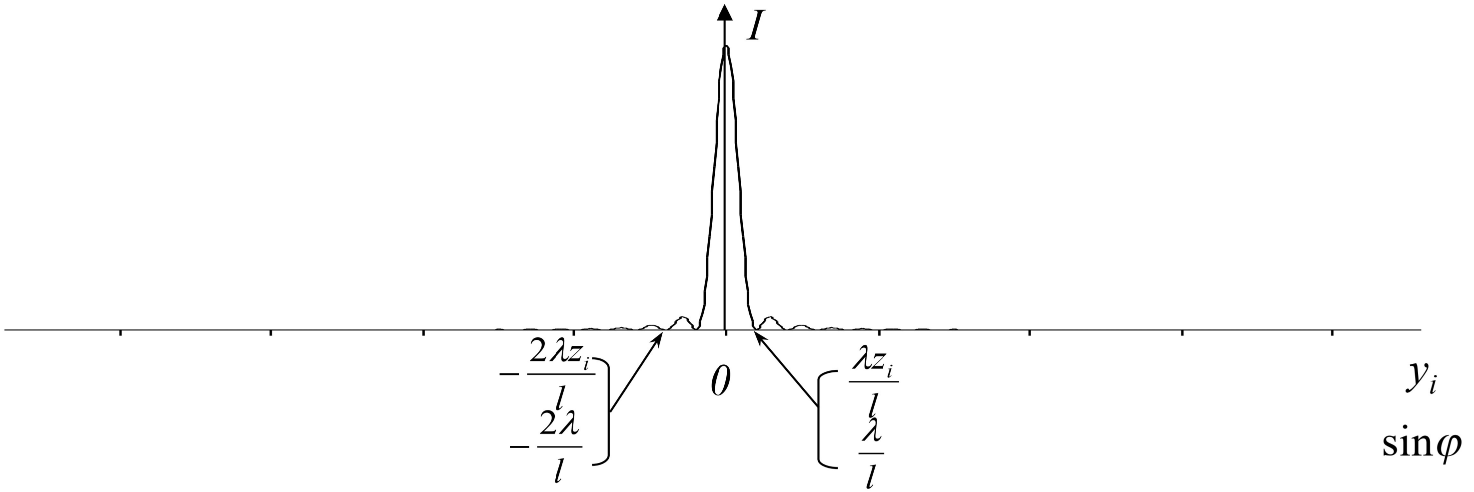
One can also write the above expression in terms of θ and φ, using (13-9):

$$I_i(\theta, \varphi) \propto \mathrm{sinc}^2(bf_x)\,\mathrm{sinc}^2(lf_y)\Big|_{f_x=\frac{k\sin\theta}{2\pi}, f_y=\frac{k\sin\varphi}{2\pi}} = \mathrm{sinc}^2\left(\frac{kb\sin\theta}{2\pi}\right)\mathrm{sinc}^2\left(\frac{kl\sin\varphi}{2\pi}\right) \quad (13\text{-}27)$$

Single-Slit Far-Field Diffraction (cont' d)

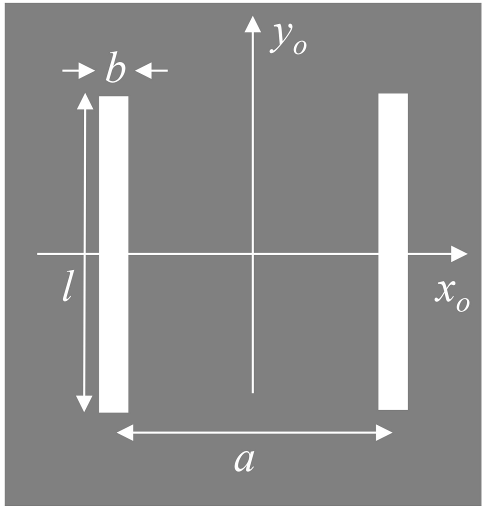
The far-field intensity plots for a single rectangle slit are given below:





Double-Slit Far-Field Diffraction

Example 13-2: Find and plot the far-field diffraction pattern of a double slit aperture as shown.



Aperture function:

$$E_o(x_o, y_o) = \Pi\left(\frac{x_o}{b}\right)\Pi\left(\frac{y_o}{l}\right) \otimes \left[\delta\left(x + \frac{a}{2}\right) + \delta\left(x - \frac{a}{2}\right)\right] \quad (13\text{-}28)$$

Far-field E field distribution:

$$E_i(x_i, y_i) \propto \mathcal{F}\left\{E_o(x_o, y_o)\right\}\Big|_{f_x = \frac{x_i}{\lambda z_i}, f_y = \frac{y_i}{\lambda z_i}} \quad (13\text{-}29)$$

$$\mathcal{F}\left\{E_o(x_o, y_o)\right\} = \mathcal{F}\left\{\Pi\left(\frac{x_o}{b}\right)\Pi\left(\frac{y_o}{l}\right) \otimes \left[\delta\left(x + \frac{a}{2}\right) + \delta\left(x - \frac{a}{2}\right)\right]\right\}$$

$$= bl\,\mathrm{sinc}(bf_x)\,\mathrm{sinc}(lf_y)\left(e^{i\pi f_x a} + e^{-i\pi f_x a}\right) = 2bl\cos(\pi f_x a)\,\mathrm{sinc}(bf_x)\,\mathrm{sinc}(lf_y) \quad (13\text{-}30)$$

Double-Slit Far-Field Diffraction (cont' d)

Far-field Intensity distribution:

$$I_i(x_i, y_i) \propto E_i^2(x_i, y_i) \propto \cos^2(\pi f_x a)\operatorname{sinc}^2(bf_x)\operatorname{sinc}^2(lf_y)\Big|_{f_x=\frac{x_i}{\lambda z_i},\, f_y=\frac{y_i}{\lambda z_i}}$$

$$= \cos^2\left(\frac{\pi a x_i}{\lambda z_i}\right)\operatorname{sinc}^2\left(\frac{bx_i}{\lambda z_i}\right)\operatorname{sinc}^2\left(\frac{ly_i}{\lambda z_i}\right) \tag{13-31}$$

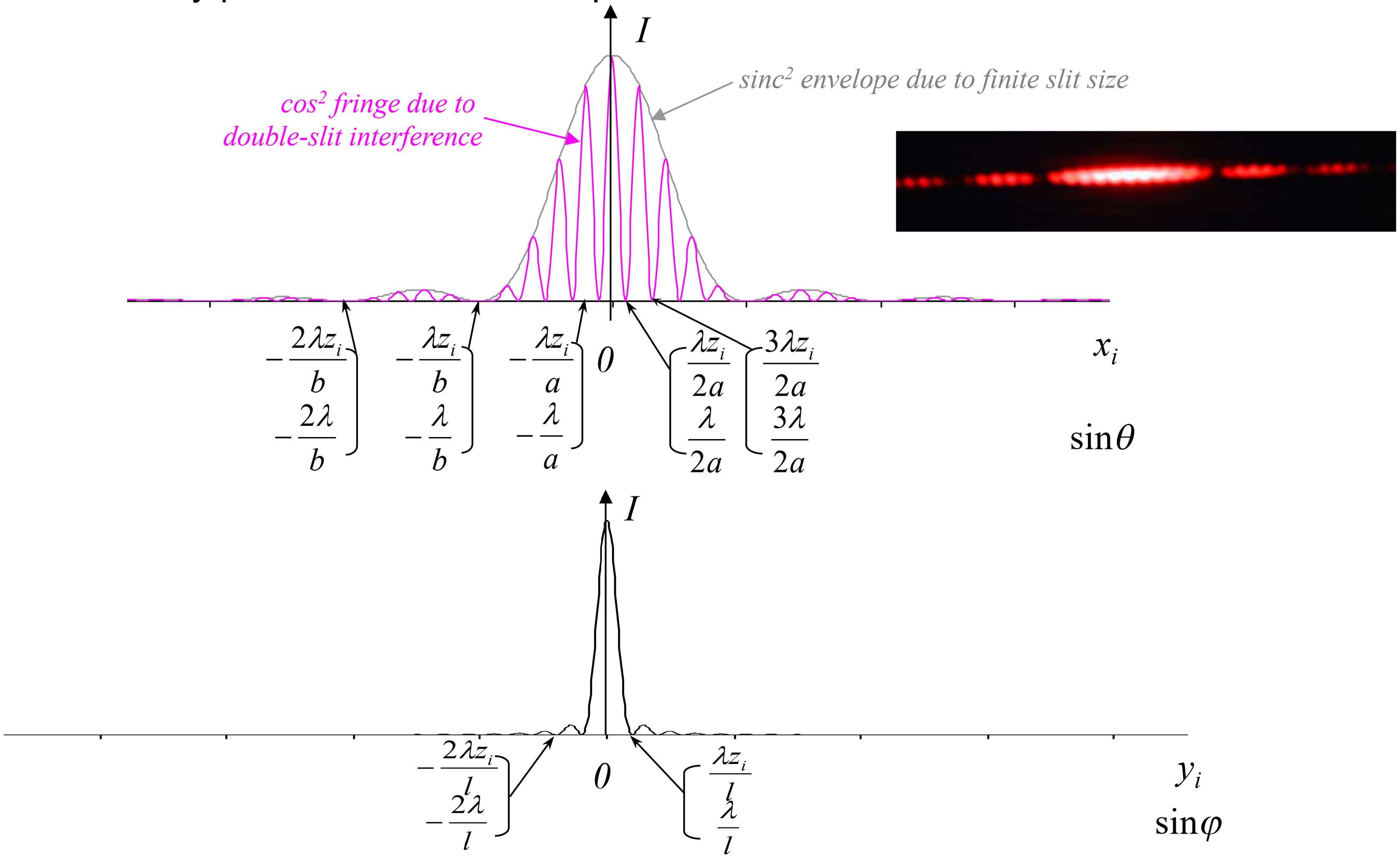
Written in terms of θ and φ:

$$I_i(\theta, \varphi) \propto \cos^2(\pi f_x a)\operatorname{sinc}^2(bf_x)\operatorname{sinc}^2(lf_y)\Big|_{f_x=\frac{k\sin\theta}{2\pi},\, f_y=\frac{k\sin\varphi}{2\pi}}$$

$$= \cos^2\left(\frac{ka\sin\theta}{2}\right)\operatorname{sinc}^2\left(\frac{kb\sin\theta}{2\pi}\right)\operatorname{sinc}^2\left(\frac{kl\sin\varphi}{2\pi}\right) \tag{13-32}$$

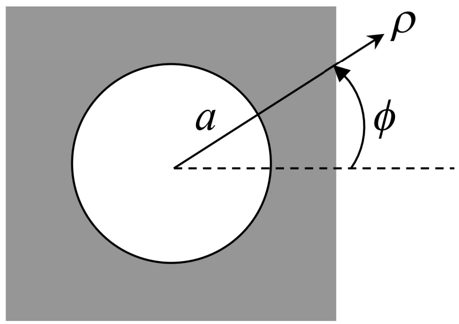
Double-Slit Far-Field Diffraction (cont' d)

The far-field intensity plots for a double-slit aperture are:



Circular Aperture Far-field Diffraction

Example 13-3: Find and plot the far-field diffraction pattern of a circular aperture as shown.



For an aperture of radius a, the aperture function is expressed as:

$$E_o(\rho_o) = \mathrm{circ}\left(\frac{\rho_o}{a}\right) \tag{13-33}$$

From (13-14), the far-field becomes:

$$E_i(\rho_i) \propto \mathcal{F}\{E_o(\rho_o)\} = \frac{1}{f_\rho}\mathrm{J}_1(2\pi f_\rho a) \tag{13-34}$$

where $$f_\rho = \frac{\rho_i}{\lambda z_i} = \frac{k\sin\theta}{2\pi} \tag{13-35}$$

Sometimes E_i is expressed in terms of the angular distance θ,

$$E_i(\theta) \propto \frac{2\pi}{k\sin\theta}\mathrm{J}_1(ka\sin\theta) \tag{13-36}$$

Far-field Circular Diffraction Pattern

The far-field intensity becomes:

$$I_i(\theta) \propto E_i^2(\theta) \propto \left(\frac{2\pi}{k\sin\theta}\right)^2 \mathrm{J}_1^2(ka\sin\theta) = 4\pi^2 a^2 \left(\frac{\mathrm{J}_1(ka\sin\theta)}{ka\sin\theta}\right)^2 \qquad (13\text{-}37)$$

The first null of I_i in Eq (13-37) occurs when

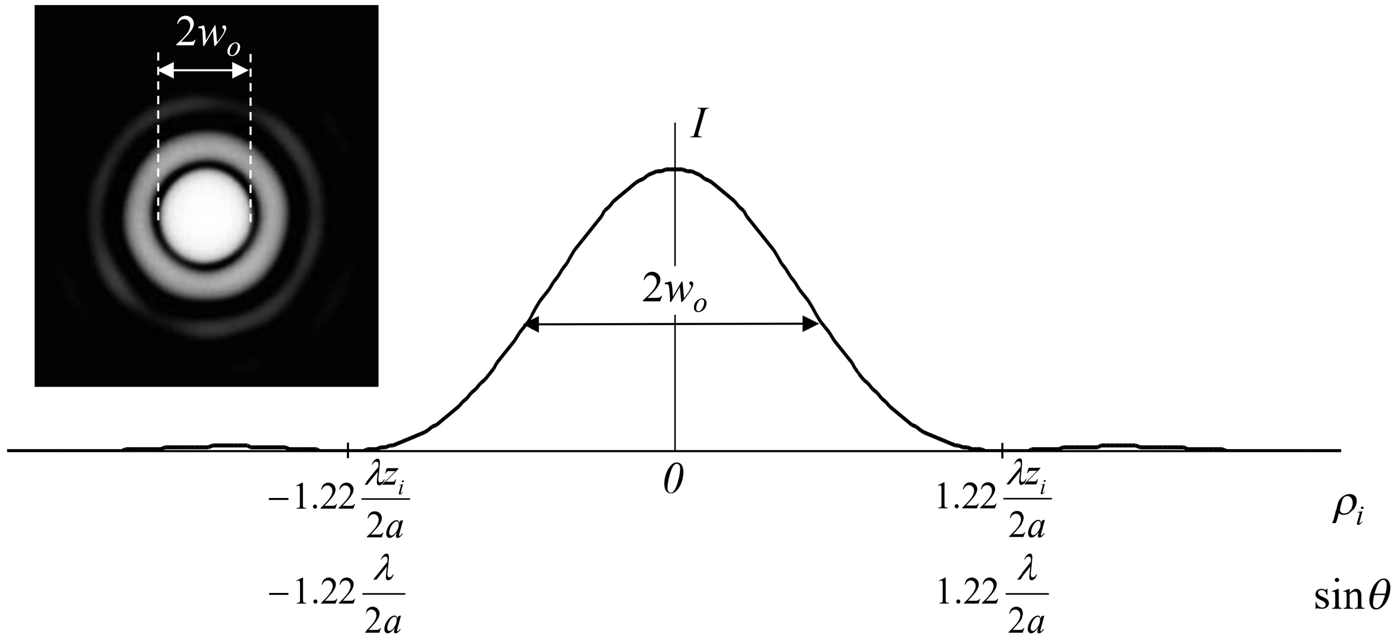
$$ka\sin\theta = 3.83 \quad \Rightarrow \quad \sin\theta = 1.22\lambda/2a \qquad (13\text{-}38)$$

or

$$2\pi \frac{\rho_i}{\lambda z_i} a = 3.83 \quad \text{or} \quad \rho_i = 1.22\lambda z_i/2a \qquad (13\text{-}39)$$

The Airy Disk

The normalized far-field intensity is plotted here:

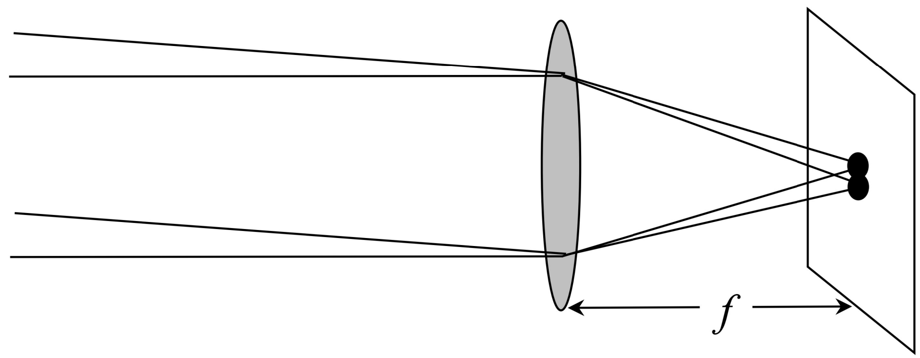


This intensity pattern is also known as the Airy Disk.

Diffraction Limit

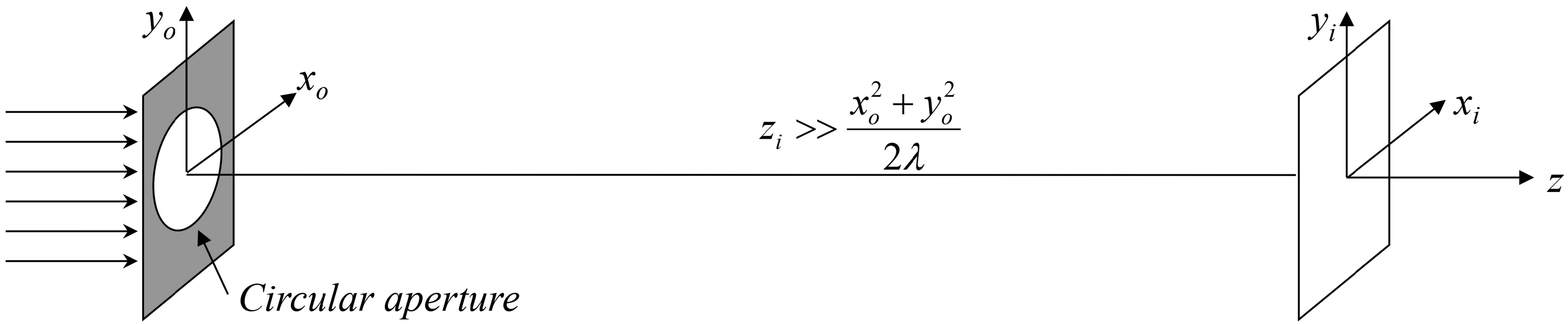
In geometric optics, we treat the focal point of a lens as a point of zero dimension. Of course, such a singularity is not physical, and it is the result of our ignoring the finite size of the wavelength.

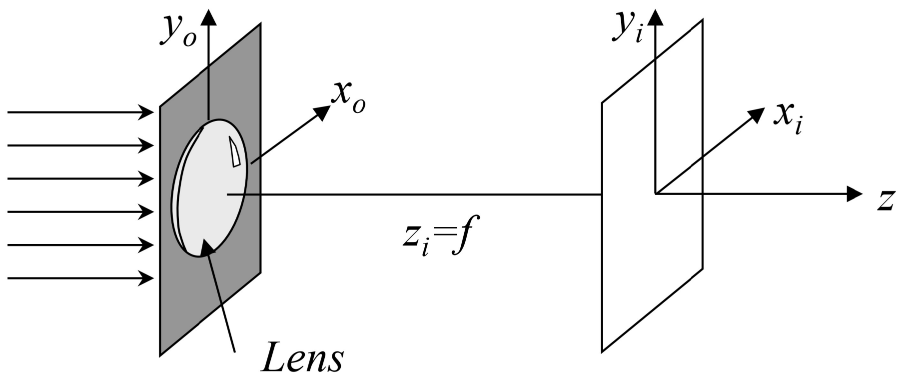
In wave optics, when the finite size of the wavelength is taken into account, collimated lightwaves can no longer be focused to an infinitely small point. Even after perfect correction of aberrations, a lens of finite aperture size has a finite focal spot size, called the diffraction limit. The diffraction limit determines the resolution of the lens. Two distant objects cannot be distinguishable on the image plane of the lens if the images they form have a separation less than the diffraction limit. (See illustration below).



Diffraction Limit (Cont' d)

In order to find the diffraction limit of a lens, we need to find the intensity pattern at the focal plane of the lens when it is illuminated by a plane wave. Recall from Page 10, the field pattern at the focal plane of a lens is equivalent to the far-field diffraction pattern, which is proportional to the Fourier Transform of the aperture function. In this case, the size of the lens determines the aperture opening.





Top: Far-field diffraction produced by a circular aperture.

Left: Far-field diffraction produced by a Lens at its focal plane.

Resolution of a Lens

Referring to the graph on page 26, we can take the size of the central lobe of the Airy Disk as the smallest focal spot (diffraction limit) a lens of finite aperture size can achieve.

Therefore, from Eq (13-39), for a lens with a focal length of f and a diameter of D, its focal spot size, $2w_o$, is approximately:

$$\boxed{2w_o \approx 1.22\lambda f / D} \qquad (13\text{-}40)$$

Correspondingly, the lens is said to have an angular resolution (aka resolving power) of:

$$\boxed{\theta_{RP} \approx 1.22\lambda / D} \qquad (13\text{-}41)$$